Simulation Report

Topic:

RAM Design and Simulation

Internship Task 2 – VLSI Design

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Tool Used: Cadence Genus, SimVision

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Chapter 1
Introduction

1.1 Introduction

Memory is an important component of digital systems used for storing and retrieving data required for processing operations. Random Access Memory (RAM) is widely used because it allows data to be accessed directly from any memory location.

Synchronous Random Access Memory (Synchronous RAM) is a type of memory that operates based on a clock signal. All read and write operations occur only at specific clock edges, ensuring reliable and synchronized data transfer. This improves stability and timing control in digital systems.

In this project, a Synchronous RAM is designed using Verilog Hardware Description Language (HDL) and verified using Cadence simulation tools. The functional behavior of the design is tested using a testbench, and simulation waveforms are analyzed using SimVision. The synthesized design is further analyzed for performance parameters such as area, power, and timing.

1.2 Theory of Synchronous RAM

Synchronous RAM is a sequential memory circuit that stores binary data in memory locations identified by addresses. The operation of the memory is controlled by a clock signal.

The main signals in synchronous RAM include:

- Clock (clk): Controls timing of operations

- Write Enable (we): Determines read or write operation
- Address (addr): Selects memory location
- Data Input (din): Data to be stored
- Data Output (dout): Stored data retrieved

When the write enable signal is high during a clock edge, data is written into the selected memory location. When write enable is low, the stored data is read from memory. The clock ensures synchronized and stable memory operation.

1.3 Objective

The primary objective of this project is to design and simulate a Synchronous Random Access Memory (RAM) using Verilog Hardware Description Language. The design is developed to perform memory read and write operations based on clock and control signals, ensuring synchronized and reliable data storage and retrieval.

The project aims to understand the working of sequential memory circuits and their implementation using hardware description languages. Functional verification of the RAM is carried out using a testbench and simulation waveform analysis to ensure correct operation under different input conditions.

Another objective is to perform synthesis of the design and analyze performance parameters such as area utilization, power consumption, and timing characteristics. The project provides practical exposure to digital memory design concepts and the VLSI design flow using Cadence simulation and synthesis tools.

1.4 Design Methodology

The Synchronous RAM is implemented using behavioral modeling in Verilog Hardware Description Language. A memory array is declared to store binary data values, and the design operates based on a clock signal.

An always block triggered at the positive edge of the clock is used to perform memory operations. When the write enable signal is active, the input data is written into the memory location specified by the address. When the write enable signal is inactive, the stored data from the selected memory location is read and assigned to the output.

A testbench is developed to verify the functional behavior of the design by applying different input combinations. The design is simulated using Cadence simulation tools, and waveform results are analyzed to confirm correct memory operation. After successful simulation, synthesis is performed to evaluate area, power, and timing performance.

Chapter 2

DESIGN METHODOLOGY

2.1 Synchronous RAM Design Implementation

The Synchronous RAM module was coded in Verilog using behavioral modeling. The design performs memory read and write operations based on clock and write enable control signals.

```
`timescale 1ns/1ps
module sync_ram (
    input clk,
    input rst, // Active high reset
    input we, // Write Enable
    input [3:0] addr, // 16 locations
    input [7:0] data_in, // 8-bit data input
    output reg [7:0] data_out // 8-bit data output
);

    // 16x8 Memory
    reg [7:0] memory [15:0];

    integer i;

    // Reset + Read/Write operation
    always @(posedge clk) begin
        if (rst) begin
            // Clear memory during reset
            for (i = 0; i < 16; i = i + 1)
                memory[i] <= 8'h00;

            data_out <= 8'h00;
        end
        else begin
            if (we) begin
                memory[addr] <= data_in; // WRITE
            end
            else begin
                data_out <= memory[addr]; // READ
            end
        end
    end

endmodule
```

Figure 2.1.1: Verilog Implementation of Synchronous RAM

Figure 2.1.1 shows the Verilog implementation of the Synchronous RAM module. The module takes clock, write enable, address, and data input signals to control memory operations. When the write enable signal is active, the input data is written into the memory location specified by the address. When the write enable signal is inactive, the stored data from the selected memory location is provided as output. The design is implemented using an always block triggered at the positive edge of the clock to ensure synchronized memory operation.

2.2 Testbench Implementation

A testbench was developed to verify the functionality of the Synchronous RAM design. It applies different input combinations to test memory read and write operations and observes the output response.

```
`timescale 1ns/1ps

module sync_ram_tb;

    reg clk;
    reg rst;
    reg we;
    reg [3:0] addr;
    reg [7:0] data_in;
    wire [7:0] data_out;

    // Instantiate RAM
    sync_ram uut (
        .clk(clk),
        .rst(rst),
        .we(we),
        .addr(addr),
        .data_in(data_in),
        .data_out(data_out)
    );

    // Clock generation (10ns period)
    always #5 clk = ~clk;

    initial begin
        clk = 0;
        rst = 1;
        we = 0;
        addr = 0;
        data_in = 0;

        // Apply Reset
        #10 rst = 0;

        // WRITE OPERATION
        #10 we = 1; addr = 4'b0001; data_in = 8'hAA;
        #10 addr = 4'b0010; data_in = 8'h55;

        // STOP WRITE
        #10 we = 0;

        // READ OPERATION
        #10 addr = 4'b0001;
        #10 addr = 4'b0010;

        #20 $finish;
    end

endmodule
```

Figure 2.2.1: Synchronous RAM Testbench Implementation

Figure 2.2.1 shows the Verilog testbench developed to verify the functionality of the Synchronous RAM design. The testbench generates clock signals and provides different input values for address, data input, and write enable signals to test memory operations. It applies input stimuli and observes the output results during simulation to ensure that the RAM correctly performs data storage and retrieval operations.

2.3 RTL / Schematic View

After synthesis, the RTL schematic was generated to visualize the hardware implementation of the Synchronous RAM design. The schematic represents internal memory elements and control logic used for read and write operations.

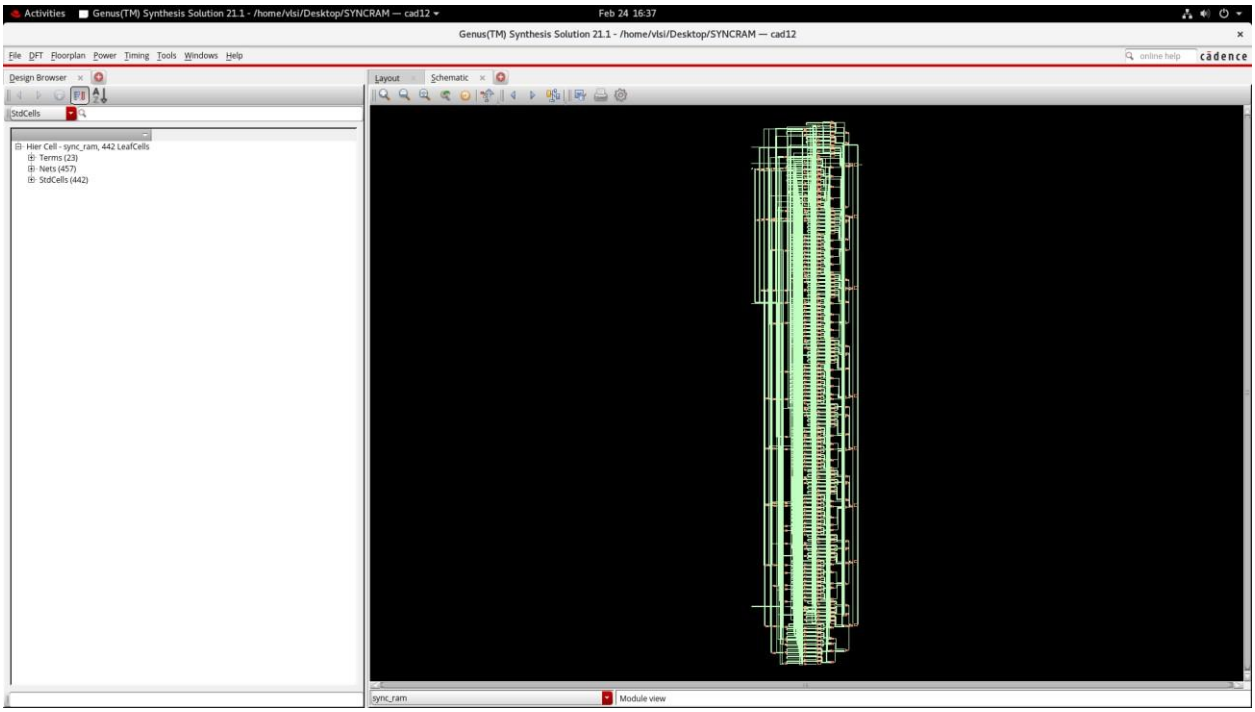

Figure 2.3.1: Synthesized RTL Schematic of Synchronous RAM

Figure 2.3.1 shows the synthesized RTL schematic of the Synchronous RAM generated after synthesis using Cadence tools. The schematic represents the internal hardware structure of the design, including memory storage elements and associated control logic. It verifies that the behavioral Verilog code has been successfully converted into a hardware-level implementation suitable for digital system integration.

Chapter 3

Simulation

3.1 Simulation Methodology

The Synchronous RAM design and testbench were simulated using Cadence simulation tools. The design was compiled and executed to verify functionality. Simulation waveforms were generated using SimVision to observe input and output signals. Different input combinations were applied, and the results were compared with expected theoretical values to verify correct memory operation.

3.2 Simulation Results and Analysis:

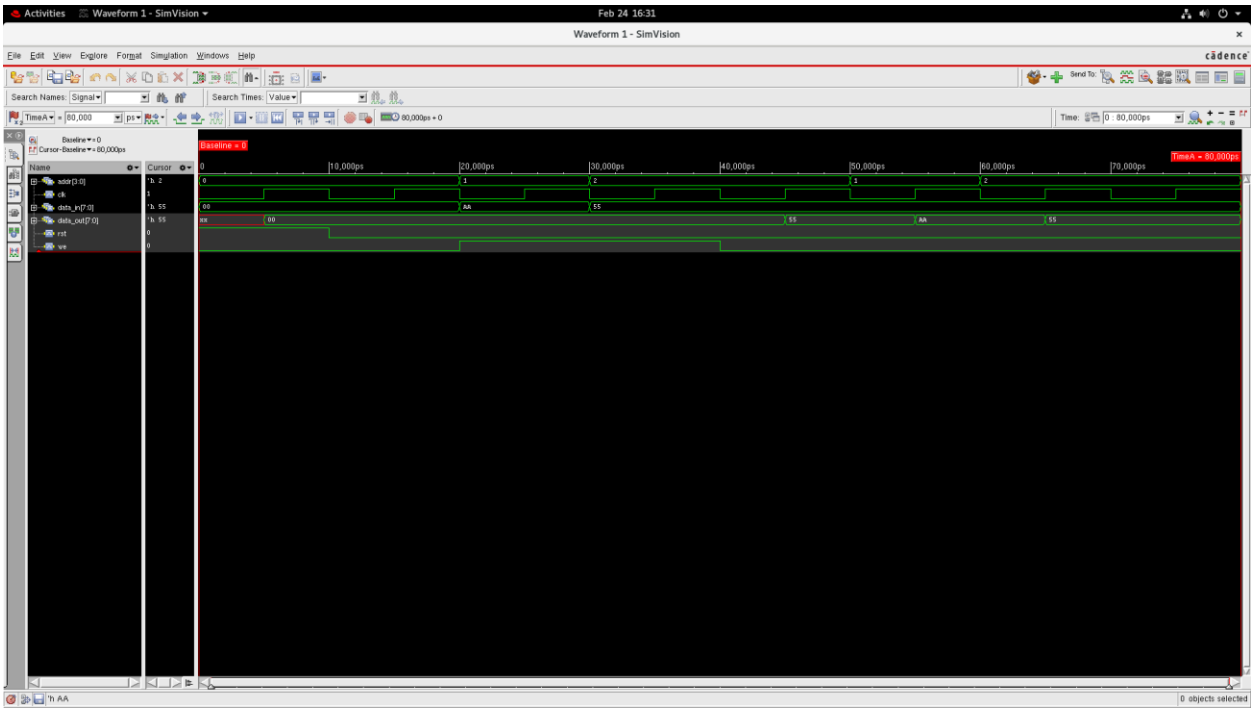

Figure 3.2.1: Simulation Waveform of Synchronous RAM

The simulation results of the Synchronous RAM were verified using the SimVision waveform viewer, as shown in Figure 3.2.1. The waveform displays the clock signal, address input, write enable signal, data input, and output data over time. During the write operation, when the write enable signal is active and a clock edge occurs, the input data is written into the selected memory location. During the read operation, the stored data is retrieved from the specified address and provided at the output.

The observed outputs match the expected theoretical results, confirming the correct functionality and operational accuracy of the Synchronous RAM design.

Chapter 4

Synthesis And Performance Analysis

After simulation, the design was synthesized using Cadence Genus to evaluate hardware performance parameters of the Synchronous RAM.

4.1 Power Analysis

The power report provides information about dynamic and leakage power consumption of the design.

```
Open ▾                    counter_power.rep                    Save
                          ~/Desktop/SYNCRAM
Instance: /sync_ram
Power Unit: W
PDB Frames: /stim#0/frame#0
  ----------------------------------------------------------------------------
   Category          Leakage     Internal    Switching        Total    Row%
  ----------------------------------------------------------------------------
     memory      0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     1.40303e-05  1.20419e-03  6.78040e-05  1.28602e-03  82.78%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     7.93162e-06  1.11549e-04  5.44222e-05  1.73903e-04  11.19%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  9.36360e-05  9.36360e-05   6.03%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ----------------------------------------------------------------------------
   Subtotal      2.19619e-05  1.31574e-03  2.15862e-04  1.55356e-03 100.00%
  Percentage            1.41%       84.69%       13.89%      100.00% 100.00%
  ----------------------------------------------------------------------------
```

Figure 4.1.1: Power Analysis Report

Figure 4.1.1 shows the power analysis report of the synthesized Synchronous RAM design generated using Cadence Genus. The report provides information about the power consumption of the circuit, including dynamic power due to switching activity and leakage power caused by static current. Power analysis is an important parameter in VLSI design as it determines the energy efficiency of the system. The results indicate that the RAM design operates with acceptable power consumption, demonstrating efficient memory implementation.

4.2 Area Analysis

The area report shows the number of logic cells and hardware resources used.

```
Activities   Text Editor                    Mar 3 10:26
Open                    counter_area.rep                    Save
~/Desktop/SYNCRAM
============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.14-s082_1
  Generated on:           Feb 24 2026  04:35:54 pm
  Module:                 sync_ram
  Technology library:     slow
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

Instance Module  Cell Count  Cell Area  Net Area   Total Area   Wireload
-----------------------------------------------------------------------------
sync_ram               442   3991.891     0.000     3991.891 <none> (D)
  (D) = wireload is default in technology library
```

Figure 4.2.1: Area Utilization Report

Figure 4.2.1 presents the area utilization report of the synthesized Synchronous RAM design. The report shows the total number of standard cells and logic elements used along with the overall area occupied by the circuit. Area analysis is a crucial factor in VLSI design because it directly affects chip size, fabrication cost, and hardware complexity. The synthesis results indicate that the RAM is implemented using an optimized number of logic cells, ensuring efficient utilization of hardware resources.

4.3 Timing Analysis

Timing analysis evaluates propagation delay and performance of the circuit.

```
Open ▾                                counter_timing.rep                                Save
                                      ~/Desktop/SYNCRAM

============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.14-s082_1
  Generated on:           Feb 24 2026  04:35:54 pm
  Module:                 sync_ram
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================


Path 1: MET (32 ps) Setup Check with Pin data_out_reg[7]/CK->D
          Group: clk
     Startpoint: (F) we
          Clock: (R) clk
       Endpoint: (R) data_out_reg[7]/D
          Clock: (R) clk

                     Capture       Launch
        Clock Edge:+    2000            0
         Drv Adjust:+      0            0
       Src Latency:+       0            0
       Net Latency:+       0 (I)        0 (I)
           Arrival:=    2000            0

             Setup:-     125
       Uncertainty:-      10
     Required Time:=    1865
      Launch Clock:-       0
       Input Delay:-     800
         Data Path:-    1033
             Slack:=      32

Exceptions/Constraints:
  input_delay             800             constraints_sdc.sdc_line_8

#-------------------------------------------------------------------------------------
#   Timing Point    Flags   Arc   Edge   Cell      Fanout Load Trans Delay Arrival Instance
#                                                    (fF)  (ps)  (ps)   (ps)  Location
#-------------------------------------------------------------------------------------
  we                -       -     F     (arrival)      3  9.7   120     0     800    (-,-)
  g8343/Y           -       A->Y  R     CLKINVX2       3  9.8    67    75     875    (-,-)
  g8338__4319/Y     -       B->Y  F     NAND2X2        5 10.2    96    85     960    (-,-)
  g8490/Y           -       AN->Y F     NAND2BX1       4  7.9   156   183    1142    (-,-)
  g8303__8246/Y     -       B->Y  R     NOR2XL         8 14.7   452   361    1503    (-,-)
  g8212__2398/Y     -       B1->Y R     AO22XL         1  1.7    71   182    1685    (-,-)
  g8062__8246/Y     -       C0->Y F     AOI221XL       1  1.6   193    52    1738    (-,-)
  g7921__2398/Y     -       C->Y  R     NAND4XL        1  1.6    94    96    1833    (-,-)
  data_out_reg[7]/D <<<     -     R     DFFQX4         1    -     -     0    1833    (-,-)
#-------------------------------------------------------------------------------------
```

Figure 4.3.1: Timing Analysis Report

Figure 4.3.1 shows the timing analysis report of the synthesized Synchronous RAM design. The timing report provides information about propagation delay, critical path delay, and overall performance of the circuit. Timing analysis is essential in digital design to ensure that the circuit operates correctly within required speed constraints. The results indicate that the design meets timing requirements, confirming reliable and stable operation.

4.4 Gate-Level Report

The gate-level report shows the number of logic gates used in implementation.

```
Open ▾                                counter_gate.rep                                Save
                                      ~/Desktop/SYNCRAM
============================================================
  Generated by:           Genus(TM) Synthesis Solution 21.14-s082_1
  Generated on:           Feb 24 2026  04:35:54 pm
  Module:                 sync_ram
  Technology library:     slow
  Operating conditions:   slow (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
============================================================

   Gate    Instances    Area   Library
-----------------------------------------
AND2X1             1     4.541    slow
AND4XL             8    54.497    slow
AO22X1            72   544.968    slow
AO22XL            56   423.864    slow
AOI221XL           8    60.552    slow
AOI222XL           8    66.607    slow
AOI22X1           21   127.159    slow
AOI22XL           19   115.049    slow
CLKAND2X2          1     5.298    slow
CLKINVX1           3     6.812    slow
CLKINVX2           1     3.784    slow
DFFQX1           120  1907.388    slow
DFFQX4             8   157.435    slow
DFFQXL             8   127.159    slow
INVXL              9    20.436    slow
NAND2BX1           7    31.790    slow
NAND2BXL           2     9.083    slow
NAND2X1            2     7.569    slow
NAND2X2            1     6.055    slow
NAND2XL           28    84.773    slow
NAND4XL            8    42.386    slow
NOR2BX1            7    31.790    slow
NOR2X1            10    37.845    slow
NOR2XL            26    78.718    slow
OAI21XL            8    36.331    slow
-----------------------------------------
total            442  3991.891

     Type      Instances   Area   Area %
-----------------------------------------
sequential           136 2191.982   54.9
inverter              13   31.033    0.8
logic                293 1768.875   44.3
physical_cells         0    0.000    0.0
-----------------------------------------
total                442 3991.891  100.0
```

Figure 4.4.1: Gate-Level Synthesis Report

Figure 4.4.1 shows the gate-level synthesis report of the Synchronous RAM design generated after synthesis. The report provides details about the number and types of logic gates used in the hardware implementation of the circuit. Gate-level analysis helps evaluate the hardware complexity and structural efficiency of the design. The results indicate that the design is implemented using an optimized number of logic gates, ensuring efficient hardware realization.

Chapter 5

Results

The simulation results clearly confirm the correct functional behavior of the Synchronous RAM for all tested conditions. The design successfully performed memory read and write operations based on the clock and write enable signals. During simulation, data was correctly stored in the specified memory locations during write operations and accurately retrieved during read operations. The waveform analysis verified that the output results matched the expected values for different input combinations.

Furthermore, synthesis analysis provided insight into the hardware performance of the design. The area report indicated efficient utilization of logic cells and hardware resources, demonstrating optimized implementation of the memory circuit. The power analysis showed acceptable dynamic and leakage power consumption, confirming energy-efficient design behavior. Timing analysis verified that the circuit met required timing constraints without violations, ensuring reliable operation.

Overall, the results validate both the functional correctness and structural efficiency of the Synchronous RAM design.

Chapter 6

Discussion

This project demonstrates the practical implementation of a memory circuit using sequential digital logic. By describing the Synchronous RAM in Verilog HDL, the behavioral functionality of the memory was successfully translated into a hardware-level implementation through synthesis. The use of clock-controlled operations ensured synchronized data storage and retrieval, providing stable and reliable memory behavior.

The verification process through simulation ensured that both read and write operations produced correct outputs under different input conditions. The waveform analysis confirmed proper synchronization of signals and correct functionality of the memory system. The synthesis results further confirmed that the design is hardware-efficient and meets performance requirements in terms of area, power, and timing.

This project reflects the standard VLSI design flow, where functional verification is followed by synthesis and performance evaluation. It strengthens understanding of sequential circuit design, memory architecture, simulation techniques, and synthesis optimization. The project also demonstrates how memory circuits are modeled using hardware description languages and implemented as physical hardware.

Chapter 7

Conclusion

The Synchronous RAM was successfully designed, implemented, simulated, and synthesized using Verilog Hardware Description Language and Cadence tools. The design correctly performs memory read and write operations based on clock and control signals, ensuring synchronized and reliable data storage and retrieval.

Functional verification through simulation waveforms confirmed accurate operation for all test cases. The synthesis analysis demonstrated efficient utilization of hardware resources, acceptable power consumption, and satisfactory timing performance. No functional or timing violations were observed, indicating stable and reliable circuit behavior.

This project provided valuable hands-on experience in sequential logic design and the VLSI development process, including coding, simulation, verification, synthesis, and performance analysis. The implemented Synchronous RAM design can be further extended to support larger memory sizes or integrated into complex digital systems such as processors and embedded applications.